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Hybrid Fuzzy Logic and Ant Colony Optimization-Based Routing for Dynamic VANET Topology Management

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***Abstract* – VANET facilitates intelligent communication between mobile and stationary vehicles through wireless connections, serving as a crucial component of ITS. A dynamic and self-organising network is established by each vehicle, which operates as a node. Enhanced vehicle safety and passenger comfort are beneficial byproducts; nevertheless, developing efficient dynamic routing protocols is challenging due to the rapid advancement of VANET architecture. This research introduces a novel hybrid model, Fuzzy-ACO-GA, to enhance routing decisions in dynamic vehicular situations. This model incorporates Fuzzy Logic, Ant Colony Optimisation, and Genetic Algorithms. The proposed methodology comprises three phases: data preparation, feature extraction, and model training. Data quality is enhanced during preprocessing by the imputation of missing values and the normalisation of features. A customised fitness function or the PSO approach is employed to determine cluster heads. Fuzzy logic enhances path optimisation and decision-making under uncertainty when utilised with ACO and GA. The experimental findings indicate that the Fuzzy-ACO-GA model attains an impressive 98.54% accuracy, far exceeding the efficacy of current methodologies. The model's capacity to adjust to frequent topological changes is validated, and its potential to enhance dynamic routing in VANETs is emphasised. This work sets a new benchmark for topology-aware routing systems in VANET.**

Keywords—Intelligent Transportation System (ITS), Unmanned Aerial Vehicle (UAV), Partial Swarm Optimization (PSO), Packet Delivery Ratio (PDR), Vehicular Ad hoc Network (VANET).

I. INTRODUCTION

Although VANET provides a foundation for innovative solutions to improve driver and passenger comfort and safety it makes it more difficult to choose and design a routing mechanism for a dynamic vehicle system that can efficiently and reliably transport packets. There are a number of routing protocols that can facilitate the transfer of packets from one node to another thereby improving the performance of VANET's dynamic system finding a trustworthy routing protocol. It provides a comprehensive analysis and explanation of topology-based routing protocols. Finding the best routing algorithms for vehicle dynamic systems is the main goal of this study [1]. Using the example of highly dense dynamic vehicle systems this article evaluates the topology-based VANET qualities in terms of performance. Software algorithms for routing based on topologies. It is done by making use of topology information to transmit data packets from their origin to their destination node. Routing procedures rely heavily on understanding the topology of individual nodes. There are essentially distinct types of this routing method: Three possible routing styles are proactive, reactive and hybrid. The goal of this article is to enhance network efficiency by focusing on topology-based routing protocols. These protocols aim to reduce network load improve end-to-end delay and increase throughput. This paper provides a concise overview of the leading topology-based routing technologies as they pertain to VANET [2]. It lays out the qualities that allow them to exchange data as well as the advantages and disadvantages of each protocol. To determine the best way to transmit data packets topology-based routing techniques make use of IP addresses in conjunction with the known network properties. In order to determine and keep the best route these protocols need topological data from the UAVs that are interacting.

Social insect populations naturally exhibit swarm-based routing behaviors such self-organization self-adaptation and cooperation to determine the best course of action. To improve traffic flow guarantee road safety and enable a wide range of ITS applications, VANET uses ad hoc communication between vehicles [3]. The use of VANETs has the potential to greatly enhance traffic management accident avoidance and driver assistance systems by enabling communication both within and between vehicles and infrastructure. However VANETs are notoriously difficult to reliably and efficiently route because of their extremely dynamic character which is a result of both the vehicles' movement and the fast changes in network topologies [4]. The urgent requirement for new routing algorithms that can respond to changes in network architecture while maintaining data integrity and timeliness is highlighted by these difficulties. In addition to considering the network's reliability in the long run these techniques need to be able to adapt to changing traffic conditions and the real-time changes that occur when vehicles enter and exit

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communication range. Performance of the VANET routing protocol was evaluated in using mobility models for real-time vehicle trajectories. Using cellular network instead of alternative communication technologies has several advantages as outlined by the authors in [5].

Here is how the rest of the paper is structured. Section II provides an overview of the associated work. In Section III, the Fuzzy-ACO-GA Model is Presented. In Section IV, they analyze and assess how well the suggested modelling approach worked. Section V concludes the whole thing.

II. Literature Survey

An intriguing area of study for the development of IT'S is VANET. Because of the constant and variable movement of cars in urban areas the topology of the network is always changing. When transporting data between vehicles in an urban setting the efficiency of routing protocols is the primary concern. Intelligent routing systems can be designed using a number of newly described methodologies including ML, AI, DL, FL and others. This strategy is driven by the fact that these technologies have the capacity and great promise to address a range of issues in ad hoc networks. The authors have introduced spectrum sensing as a classification problem and suggested a RNN based sensing approach using the LSTM layer [6]. The LSTM network is trained using a variety of signal noise data types with the power spectrum of the input signal serving as the input. The noise class's confidence level determines the decision. Because it is a DL based approach the network can adapt to varying degrees of noise and learn the parameters automatically. The continual monitoring of these systems is essential to their effective operation which in turn opens the way to applying ML algorithms to the massive amounts of data produced by various VANET applications [7]. The goal of ML is to enable systems to learn and improve themselves autonomously by analyzing and interpreting data that has already been processed.

The goal of VANET is efficiently implemented by these algorithms which enable supervised and unsupervised learning on this acquired data. One potential option is Q-learning one of the most popular and user-friendly RL approaches [8]. The ITS agent may make the best decisions possible without knowing anything beforehand by modifying their actions based on the reward function of environmental feedback. Using the node's mobility and transmitting power as inputs and the Hello interval as outputs presented an ANN-based technique to calculate SI. While network latency increases throughput and PDR improve. There have been numerous attempts to apply the principles of SDN to vehicle networks so that these networks can take use of the programmability and flexibility that SDN provide. Improving VANET performance and making it more suited for particular services and applications is the ultimate goal of this adaptation [9]. To meet the increasing demands of future networks SDN are being used in data centers and operator networks among other revolutionary new concepts in networking. Among SDN's characteristics include network programmability open interfaces and centralized management control. It optimizes the fitness function in both the DLSR and DLC protocols by designing a DNN model. While DLC takes into account cosine distance, cosine similarity and the node's remaining energy DLSR takes into account characteristics like distance hop count and remaining energy. One of the most effective ML methods and DL uses DNN to examine intricate data patterns [10]. When compared to more traditional approaches this one has more benefits particularly when it comes to determining what characteristics differentiate benign from harmful network activity. The DL improves detection accuracy by efficiently processing high-dimensional data. By improving the optimization speed and utilizing DL to choose the best path based on fitness function values it may actively avoid black hole nodes as stated in [11]. This would reduce delays and enhance network performance.

Through the integration of a ACO, GA, and fuzzy logic, this paper proposes a hybrid approach called Fuzzy-ACO GA. The goal is to build a Dynamic Routing VANET that strikes a good balance between interpretability and accuracy. For the VANET Dynamic Routing Topology problem and its outcomes, Fuzzy-ACO-GA will be demonstrated to be superior and applicable.

III. Proposed System

Networking among unmanned aerial vehicles (UAV), often called flying ad hoc networks, has evolved rapidly in response to the ever-increasing demand for flexible and portable communications. Nevertheless, FANET routing is difficult because UAVs have unique properties including frequent topology change, high mobility, and 3D space movement. These features make it very challenging to create new FANET routing protocols.

This dataset includes information for simulating false information attacks on urban and highway VANETs. The vehicle IDs are displayed in the first column of the CSV files, while the corresponding speeds in kph are shown in the second column. Depending on the scenario, honest vehicles can reach speeds of 80–110 kph on highways and 45–65 kph in urban areas [12]. Vehicles that exceed these speeds are considered to be malevolent. The naming convention is followed by the files: scenario_percentage of harmful vehicles.csv.

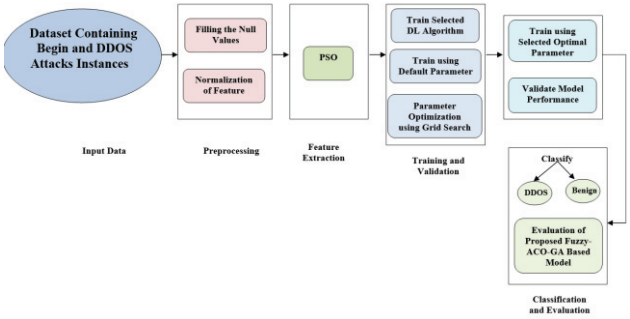

Fig. 1. Overall Proposed Model Diagram

They may see the whole suggested system in Figure 1. After receiving the input data, it undergoes preprocessing, then moves on to feature extraction, and finally partitions the dataset into training and testing. Model evaluation and classication performance follow [13].

A. Preprocessing:

1) Filling the Null Values:

Sensor failure and outside interference are two potential causes of features in the car having null values. They present a method for filling in missing values that relies on assessments of spatial and temporal similarity.

When comparing car features in the same traffic situation, spatial similarity is a useful metric to use. Equation (1)

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displays the similarity between e_r and e_q at time t based on the PCC [14].

$$H_{rq}(g) = \frac{m\sum_{p=1}^{m} u_r^p(g)u_q^p(g) - \sum_{p=1}^{m} u_r^p(g)u_q^p(g)}{\sqrt{m\sum_{p=1}^{m}\left(u_r^p(g)\right)^2 - \sum_{p=1}^{m}\left(u_r^p(g)\right)^2}\sqrt{m\sum_{p=1}^{m}\left(u_r^p(g)\right)^2 - \sum_{p=1}^{m}\left(u_r^p(g)\right)^2}} \quad (1)$$

$u_r^p(g)$ is the $p-th$ vehicle feature of e_r, while $u_q^p(g)$ is the $p-th$ vehicle feature of e_q at g, and m is the vehicle feature dimension.

Comparing how different aspects of a vehicle change over time is what temporal similarity is all about. In the case of a vehicle node e_r the likelihood of a feature being identical increases as the time interval decreases. Therefore, Equation (2) shows the temporal similarity of e_r at time g.

$$G_r(g) = v^{-\alpha g} \quad (2)$$

The natural constant v and the temporal factor α are defined. The value for the null value of the $p-th$ vehicle feature of e_r is shown in Equation (3), which is based on spatial similarity and temporal one.

$$u_r^p(g) = H_{rq}(g) \times u_q^p(g) + G_r(g) \times u_r^p(g - \Delta g) \quad (3)$$

Normalization of Feature:

They normalise vehicle features into the same interval so that the dimensions of the features can be unified in Equation (4).

$$c_r^p(g) = \frac{u_r^p(g) - u_r^{p,min}}{u_r^{p,max} - u_r^{p,min}} \quad (4)$$

where $u_r^{p,min}$ is the minimum value of the $p-th$ vehicle feature and $u_r^{p,max}$ is the maximum value of the $p-th$ vehicle feature, and $u_r^p(g)$ is the $p-th$ vehicle feature of e_r at g.

B. Feature Extraction:

1) PSO Method:

As with other swarm intelligence methods, the PSO method falls into that category. Optimisation difficulties are frequently addressed using this approach. A population-based search algorithm, the PSO algorithm takes its cues from the cooperative actions of flocks of birds. Everyone is a potential answer, and they all stand in for particles. The hyperdimensional search space employs particles fly operating at a speed that can be adjusted. Each particle's location and speed within the swarm dynamically alter based on its own unique light experience and the people' socio-psychological tendencies. Indeed, with the help of their memories, particles can commit the optimal location in the search space to memory. Experience or knowledge impacts this change [15]. This method is employed to regain the premier positions in the search space and obtain the optimal solution.

The PSO approach has numerous advantages, including:

- The comprehension and execution of the procedure are straightforward.
- Few parameters are available for adjustment.
- Computing this method does not require a lot of time.
- Velocity, not merely position, is the determining factor in the process.
- Rapid convergence is possible.

C. Model Training:

1) Learning Process of GA for the DB:

They utilise a genetic approach for the construction of the DB, which comprises three standard components: scaling factors, definitions of membership functions, and the quantity of labels per variable [16].

The genetic process commences with a population of solutions known as chromosomes, which form the initial generation $(T(0))$, and subsequently undergoes evolution. Applying a set of genetic operators on the people of generation $T(g)$, a new population $(T(g+1))$ is formed while a specified termination condition is not fulfilled. Each chromosome is evaluated using an evaluation function.

2) The Rules of Canditate Generating:

Two methods exist for generating potential rules: example-based guidance and fuzzy grid-based guidance.

Following the example-based linguistic rule-generation method, each rule in the dataset is derived from a distinct example. The input-output data set's optimal guideline for handling each example will shape this. To obtain the structure of each rule, our team select a specific example (v_o) and assign the linguistic label corresponding to the fuzzy set that encompasses every component of $Z_r^o \in Z_r$ with $(Z_1^o, .., Z_m^o, Y^o)$ and $Y_o \in Y$ to each rule variable.

To get a rule for each group that considers all of the cases, another way to construct the language rules is to use a fuzzy grid to bracket the examples. To get the fuzzy grid, for each input variable $Z_1 \times, .., Z_m$ through, take the Cartesian product of the language words in the principal fuzzy partition. With respect to every m −dimensional fuzzy subspace input (H_h), execute:

- Assemble the set of, V_h' from the input-output data pairs found in this subspace using Equation (5)

$$V_h' = \left\{v_{o^h} = \left(c_1^{o^h}, .., c_m^{o^h}, b^{o_h}\right) \in V such\ that \tau_{Z_1^H}\left(c_1^{o^h}\right), .., \tau_{Z_m^H}\left(c_m^{o^h}\right) \neq 0\right\} \quad (5)$$

- If the absolute value of $|V_h'| \neq 0$, meaning that this space zone contains no data, then shown in Equation (6):

$$Y^h = \left\{Y_p \in Y, p \in \{1, .., |Y|\}\right\} such\ that \exists v_{o^h} \in H \quad (6)$$

Let $V_h' with \tau_{Y_p}(b^{o_h}) \neq 0$ denote the collection of linguistic labels that comprise the term set of the output variable and serve as examples.

- Create rules for each subspace (Y^h) along with their potential outcomes. Integrate the rule into Equation (7).

$$I_{p^h}^H = IF\ C_1 is\ Z_1^h and, .. and\ C_m is Z_m^h Then\ B\ is\ Y_{p^h}^H \quad (7)$$

Algorithm 1 displays the generic scheme of the Fuzzy-ACO-GA technique.

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Algorithm 1: Feature Selection		
1.	Use the GA to generate the database.	
2.	For every subspace fuzzy input, come up with the potential rules.	
3.	In each subspace, choose the rule that promotes cooperation the most. This process is carried out in two stages:	
	3.1	For any subspace H that contains instances, get the set of all potential consequents (Y^h).To accomplish this, we can use the formula $X = \left\{i_q \in \{1,..,M_h\}\right\}$ $such\ that Y^{i_q} \neq \varphi$ where $M_V = \lvert V \rvert$ is the number of example-containing subspaces and $q \in \{1,..,M_V\}$ is the collection of identifications of those candidate sets of consequent terms that contain an element.
	3.2	To get the most accurate combination $\left\{Y^{i_1}_{p^{i_1}},..,Y^{q}_{p^{i_1}},...,Y^{i_{M_V}}_{p^{i_{M_V}}}\right\}$, run the ant colony optimisation method. A random combination generator is used to obtain the initial solution. The neighbour technique chooses a certain $i_q \in V$ at random so that $v_{i_q} \geq 2$ and replaces it with $Y^{i_q}_{p^{i_q}}$ that is generated$\left\{1,..,V_{i_q}\right\}$at random in $Y^{i_q}_{p^{i_{q^1}}}$ with $p^{i_{q^1}}$ so that $p^{i_{q^1}} \neq p^{i_q}$. Each solution is assessed based on an index that measures the degree of collaboration of the encoded rule set. MSE is a global error function that the algorithm utilises in this scenario. It is defined as $$MSE = \frac{1}{2.M}\sum_{o=1}^{M}(B^o - b^o)^2$$ By substituting the known desired output B^o for the output b^o produced from the FRBS when the example v^o is utilised. Rule cooperation improves when the measure approaches zero, indicating higher global performance.
4.	Find the value of this fitness metric $$U_v = \omega_1.MSE + \omega_2.\#I$$ considering the weighting percentages ω_1 and ω_2 the number$(\omega_1 + \omega_2 = 1)$ of rules $\#I$.To get more compact linguistic models, taking $\#I$ in U_v will penalise the FRBS by the number of rules.	
5.	Return to Step 1 if the Termination Parameter is Unmet. In any case, the produced KB is ideal.	

3) Fuzzy ACO Rules Learn to Linguistic:

The following is the fundamental method of operation for ACO algorithms: An ant colony of a fixed size builds an instance-specific graph track by instance in accordance with a probabilistic transition rule that is information-dependent at each iteration. As a result, the pheromone trails are refreshed. In order to achieve this, we must first reduce them by a constant factor before they can strengthen the qualities of the built solutions. The rule that develops this task is the global pheromone trail update.

When using ACO to find the optimal set of cooperative rules, it helps to think of the problem as a weighted graph representation of a combinatorial optimisation problem. One can tackle the problem by presuming a predetermined quantity of subspaces and perceiving learning as the assignment of output fuzzy partition labels to these subspaces based on an optimality criterion. This is ultimately an assignment problem. It can establish a parallel between subspaces and locations, as well as between facilities and capacities.

Assigning a consequent to more than one subspace is theoretically possible, and the possible consequents for each subspace can differ. Since one assignment does not limit the following ones, we can infer from these features that the order in which each subspace is chosen to receive a subsequent is not important.

IV. Result and Discussion

A subtype of multi-hop Mobile Ad hoc Networks, a Vehicular Ad hoc Network (VANET) enables wireless interfaces in automobiles to communicate with one another and with stationary equipment situated along urban roadways or highways. Routing protocols' effectiveness in a real VANET is significantly affected by the dynamic features of vehicle mobility, including rapid speed and predictable, limited mobility patterns. The findings of the analysis and simulation show that VANET is characterised by a truncated scale-free degree distribution with a power-law distribution, and it has small world network characteristics.

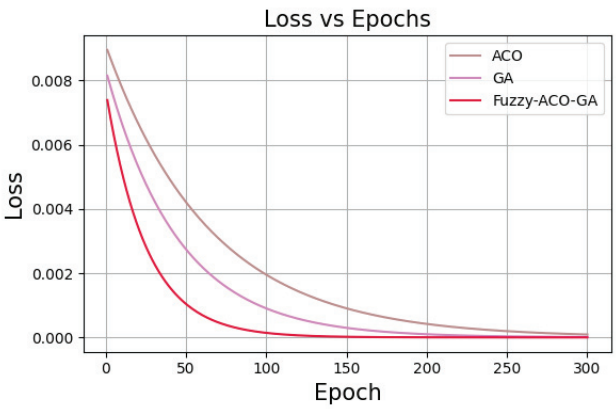

Fig. 2. All Models Loss Curve

Furthermore, as illustrated in Figure 2, the Fuzzy-ACO-GA exhibits the minimal loss. The Fuzzy-ACO-GA model stands out from the rest as the most efficient one.

TABLE I. Performance Comparison(%)

Models	**Precision**	**Recall**	**F1-Score**
ACO	93.42	91.39	95.39
Fuzzy-ACO-GA	**96.61**	**94.12**	**98.76**
GA	90.30	88.42	92.52

Table 1 displays the results of the experiment, as discussed earlier. The table provides a concise summary of the three models' overall performance based on F1 score, recall, and precision. Our proposed Fuzzy-ACO-GA model beats out ACO and GA, according to the data in the table.

Figure 3 further shows that the Fuzzy-ACO-GA has the best accuracy. The Fuzzy-ACO-GA model stands out from the rest as the most efficient one.

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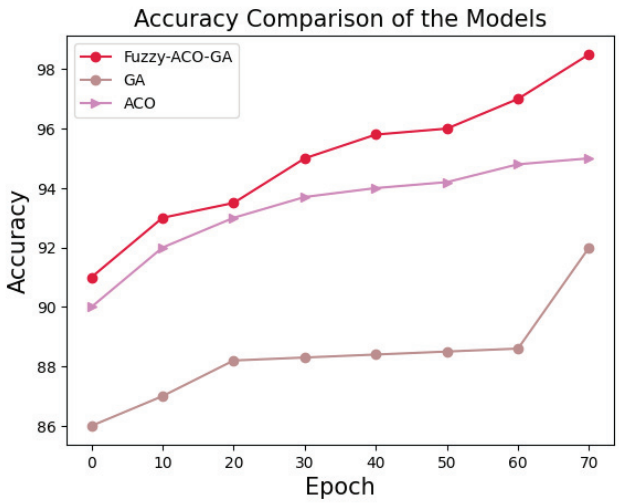

Fig. 3. Accuracy Comparison

TABLE II. COMPUTATION TIME AND ACCURACY

Models	Accuracy	Processing Time
ACO	95.18	26.87
Fuzzy-ACO-GA	**98.54**	**18.54**
GA	92.29	33.76

Table 2 shows the results of our method, including the accuracy and processing times of Fuzzy-ACO-GA for different models.

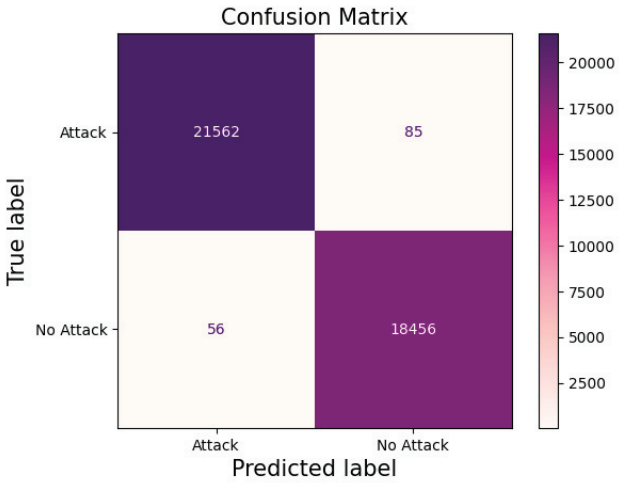

Fig. 4. Confusion Matrix

Figure 4 shows that a small improvement in accuracy to 98.54% was accomplished by further optimizing the Fuzzy-ACO-GA parameters using varied ranges and set values.

Figure 5 shows a line graph representing the False Positive Rate (FPR) of four machine learning techniques. A key indicator in intrusion detection systems (IDS) is the false positive rate (FPR), which shows how often harmless actions are mistakenly marked as threats.

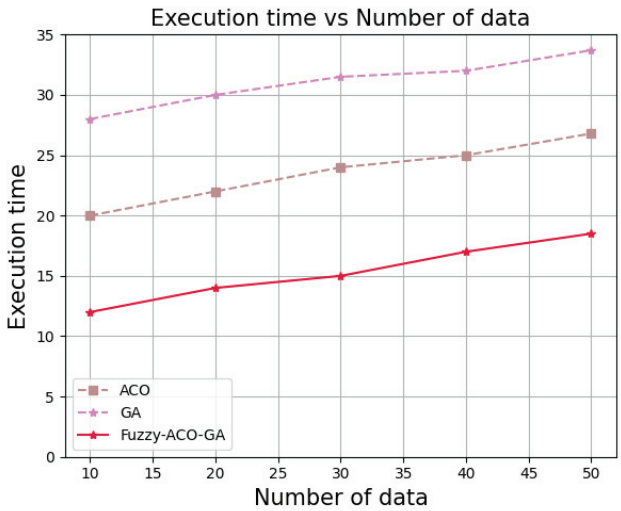

Fig. 5. Execution Time for all the Model

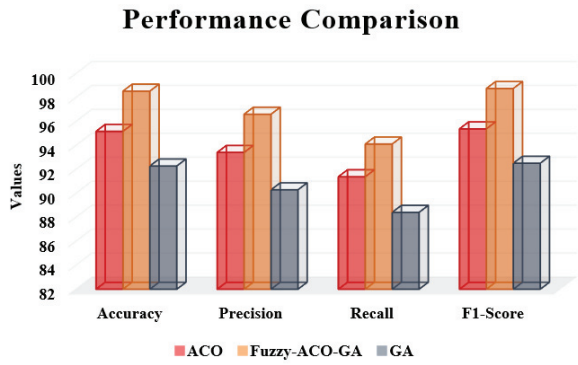

Fig. 6. Performance Comparison

Figure 6 displays a comparison of all the machine learning techniques utilised in this paper. The results demonstrate that the Fuzzy-ACO-GA methodology outperforms all the others across all evaluation parameters.

V. CONCLUSION

VANET employs ad hoc communication among vehicles to enhance traffic flow, ensure road safety, and facilitate diverse Intelligent Transportation System applications. Traffic management, accident prevention, and driver assistance systems can all be greatly improved with the help of VANETs, which allow communication not just between vehicles but also between vehicles and infrastructure. On the other hand, VANETs are notoriously difficult to implement efficient and dependable routing for because of how dynamic they are, caused by things like vehicle movement and constantly changing network topologies. In data preprocessing, features are normalised and null values are filled in to provide high-quality data. As a first step, either the cluster heads are determined using the PSO algorithm or a new fitness function is constructed to find the optimal solutions. This research introduces a novel approach to building VANET Dynamic Routing systems: Fuzzy-ACO-GA, a hybrid methodology that combines ACO, GA, and fuzzy logic. According to the results, Fuzzy-ACO-GA outperforms the other methods. When compared to other models, Fuzzy-ACO-GA provides the greatest accuracy rate of 98.54%.

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